CURRICULUM VITAE · ACADEMIC

Yi Xia *(Sean)*

PhD Candidate in Computer Vision & Human Activity Recognition · Auckland University of Technology

Tāmaki Makaurau · Auckland · Aotearoa New Zealand
+64 27 368 7961
xiayi.shawn.001@gmail.com
Google Scholar · AUT Profile · LinkedIn

70+	3	Q1
CITATIONS	H-INDEX	PAT. RECOG.

RESEARCH STATEMENT

I am a final-year PhD candidate at the **Auckland University of Technology (AUT)**, working at the intersection of **deep learning**, **computer vision**, and **graph neural networks**. My doctoral research designs *frequency-aware spatio-temporal topology learning* for skeleton-based human activity recognition (HAR), with a recent first-author paper accepted at *Pattern Recognition* (Q1, IF = 7.6). My earlier master's research on real-time object detection and tracking in agricultural video — the KiwiDetector / KiwiTracker line of work — has accumulated **50+ citations on a single paper**. I am building a research programme around *practical, interpretable, and computationally efficient* deep models for video and structured time-series data. Four years of parallel industry AI engineering at TOHU MEDIA in Auckland reinforce a deliberate stance: research that ships.

RESEARCH INTERESTS

- Skeleton-based human activity recognition
- Real-time object detection & tracking
- Frequency-domain representation learning
- Generative AI & agentic systems for vision workflows
- Spatio-temporal graph convolutional networks
- Video analytics for agriculture & media
- Attention & Transformer architectures for vision
- Practical, deployable deep-learning systems

EDUCATION

Doctor of Philosophy (PhD), Computer & Information Sciences — *Auckland University of Technology*

Oct 2023 – Oct 2026 (expected)

Faculty of Design and Creative Technologies, School of Engineering, Computer & Mathematical Sciences. Member, Institute of Robotics & Vision (IoRV).

Provisional thesis: *Deep Learning for Skeleton-Based Human Activity Recognition.*

Supervisors: **Dr Raymond Lutui** (primary, AUT) · **Assoc. Prof. Sira Yongchareon** (AUT).

Funded by the **AUT PhD Full Scholarship** (tuition + living stipend, Oct 2023 – Oct 2026).

Master of Computer & Information Sciences (Computer Vision) — *Auckland University of Technology*

2021 – 2023

Research master. Thesis: *Kiwifruit Detection and Tracking from a Deep-Learning Perspective Using Digital Videos.*

Supervisors: Dr Minh Nguyen (primary) · Prof. Wei Qi Yan.

Bachelor of Engineering, Logistics Engineering (Software Development) — *Dalian Neusoft University of Information, China* 2011 – 2015

PUBLICATIONS

*Reverse chronological. Names in **bold** indicate first author. Citation counts as of April 2026; live counts on Google Scholar.*

JOURNAL ARTICLES

01 **Xia, Y.**, Lutui, R., Yongchareon, S., & Sheng, Q. Z. (2026). Frequency-Aware Spatio-Temporal Topology Learning for Skeleton-Based Human Activity Recognition. *Pattern Recognition*. Q1 · IF = 7.6

REFEREED CONFERENCE PAPERS

01 **Xia, Y.**, Lutui, R., Yongchareon, S., & Sheng, Q. Z. (2025). Dynamic Self-Attention Gated Spatial-Temporal Graph Convolutional Network for Skeleton-Based Human Activity Recognition. In *Proceedings of the 31st International Conference on Neural Information Processing (ICONIP 2024)*, Springer LNCS.

02 **Xia, Y.**, Nguyen, M., Lutui, R., & Yan, W. Q. (2023). Multiscale Kiwifruit Detection from Digital Images. In *Proceedings of the Pacific-Rim Symposium on Image and Video Technology (PSIVT 2023)*, Springer. 7 cites

03 **Xia, Y.**, Nguyen, M., & Yan, W. Q. (2023). Kiwifruit Counting Using KiwiDetector and KiwiTracker. In *Proceedings of the SAI Intelligent Systems Conference 2023*, Springer Lecture Notes in Networks and Systems. 11 cites

04 **Xia, Y.**, Nguyen, M., & Yan, W. Q. (2022). A Real-Time Kiwifruit Detection Based on Improved YOLOv7. In *Proceedings of the 37th International Conference on Image and Vision Computing New Zealand (IVCNZ 2022)*, Springer LNCS. 50 cites

THESES

01 **Xia, Y.** (2023). *Kiwifruit Detection and Tracking from a Deep-Learning Perspective Using Digital Videos*. *Master's thesis*, Auckland University of Technology.

CONFERENCE PRESENTATIONS

- "Dynamic Self-Attention Gated Spatial-Temporal GCN for HAR," oral presentation, *ICONIP 2024*.
- "Kiwifruit Counting Using KiwiDetector and KiwiTracker," presentation, *SAI Intelligent Systems Conference*, 2023.
- "A Real-Time Kiwifruit Detection Based on Improved YOLOv7," oral presentation, *IVCNZ 2022*, Auckland, October 2022.

RESEARCH EXPERIENCE

Doctoral Researcher — *Institute of Robotics & Vision (IoRV), AUT* Oct 2023 – present

- Active member of the IoRV research group, contributing to its programme in deep learning, computer vision, and skeleton-based human activity recognition.
- Designed and implemented graph-convolutional architectures (FATL-GCN, Dynamic Self-Attention Gated ST-GCN) for skeleton-based HAR; trained and evaluated on NTU RGB+D 60/120, Kinetics-Skeleton and related benchmarks.

- Published in *Pattern Recognition* (2026, Q1) and ICONIP (2024); two further first-author papers in preparation.

Research Assistant *— School of Engineering, Computer & Mathematical Sciences, AUT* Nov 2022 – Jul 2023 (multiple appointments)

- AUT Summer Research Assistant (Nov 2022 – Feb 2023) under the AUT Summer RA Scholarship.
- Research Assistant (Jun – Jul 2023) under the SECMS Research Assistant Fund.
- Worked on the kiwifruit detection / counting research line, co-developing KiwiDetector and KiwiTracker; contributed to three peer-reviewed publications during these appointments.

INDUSTRY EXPERIENCE

Artificial Intelligence Developer *— TOHU MEDIA · Tāmaki Makaurau / Auckland* Feb 2022 – present

Māori-owned media-production company. Full-time since June 2025; previously part-time alongside MSc and PhD studies.

- Responsible for the company's AI capability end-to-end. Design and ship **object-detection and OCR systems** that operate on client video footage inside internal production pipelines (specific deliverables under NDA).
- Engineer for the messy edge cases of real footage that off-the-shelf CV models stumble on: variable cinematography and framing, dense overlay text in non-standard layouts, archival material with degraded contrast, and long-form takes where naive per-frame inference doesn't fit a single-workstation GPU budget.
- Own the full ML lifecycle: dataset scope and labelling spec, training and evaluation in PyTorch, **Docker-packaged inference deployed to the company's cloud GPU servers**, and iteration with editors and the creative director on what "useful output" looks like inside an actual editing workflow.
- Demonstrates ability to translate research methods into production constraints and to communicate technical work with non-academic, creative-industry stakeholders.

Tech Service Specialist (part-time) *— PB Tech, Auckland* Apr – Dec 2023

- Diagnosed and repaired hardware and software issues, and supported walk-in customers across consumer and SMB IT.

TEACHING EXPERIENCE

Teaching Assistant *— School of Engineering, Computer & Mathematical Sciences, AUT* 2022 – present

- TA on multiple undergraduate computer-science papers (programming, computer vision, AI).
- Lab leadership, marking, consultation hours, and mentoring of students on assignments and final-year projects.

Co-Founder — *Baiyin Pinuo Training School, China* 2019 – 2021

- Co-founded a private supplementary-education school; designed curriculum and led day-to-day academic operations. Divested in 2021 to relocate to New Zealand for postgraduate study; the school remains in operation under my co-founder.

Junior Lecturer, Department of Information Engineering — *Baiyin Vocational & Technology College, China* Aug 2015 – May 2019

- Taught Computer Fundamentals, Java programming, IoT applications and RFID technologies to ~120 students per year over four academic years.
- Course design, lecture delivery, lab supervision, examinations and pastoral care.

AWARDS & SCHOLARSHIPS

- **AUT PhD Full Scholarship** — tuition + living stipend (Oct 2023 – Oct 2026).
- **SECMS Research Assistant Fund** — May 2023, AUT.
- **AUT Summer Research Assistant Scholarship** — December 2022.

PROFESSIONAL SERVICE & MEMBERSHIPS

- Member, AUT Institute of Robotics & Vision (IoRV), 2022 – present.
- Reviewer / sub-reviewer for international conferences in computer vision and AI (details on request).

TECHNICAL SKILLS

Programming Python (expert · 8+ years) · Java (taught for 4 years).

Deep-learning frameworks PyTorch (primary) · TensorFlow / Keras · Hugging Face.

CV & scientific computing OpenCV · scikit-learn · NumPy · pandas · SciPy · Matplotlib.

Methods CNNs · GCNs · Transformers / attention · object detection (YOLO family) · object tracking · action recognition · OCR · semantic / instance segmentation · skeleton-based HAR · frequency-domain analysis.

Generative AI & LLM Claude / GPT API integration · agentic workflows · agent skills · tool-use & function calling · prompt engineering.

Deployment Docker · containerized model inference on cloud-hosted GPU servers.

Tooling Git / GitHub · Linux · CUDA · Jupyter.

LANGUAGES

Mandarin Chinese
Native

English
Professional working proficiency

REFEREES

Available on request. Referees include the doctoral supervisory team (**Dr Raymond Lutui**, primary; **Assoc. Prof. Sira Yongchareon**, second supervisor — both AUT) and an industry referee from TOHU MEDIA.